Financial Fraud Transaction Analysis

Detecting risk patterns across 10,000 transactions — Data Analytics Portfolio Project

[Your Name] | [Date] · Tools: SQL · Python · Power BI

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Agenda

Overview

Problem, dataset, and objectives

Methodology

SQL → Python → Power BI analytics pipeline

Findings

Key fraud patterns and quantitative signals

Dashboard

Live monitoring & filtering

Recommendations

Operational controls and projected impact

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Problem Statement

Financial fraud imposes material losses and operational burden. Manual review is slow, inconsistent, and costly. We need a data-driven system to identify high-risk transactions in near real time and prioritize investigation effort.

Goal

Uncover risk patterns and enable proactive prevention

Scope

10,000 transactions — exploratory analysis and dashboard prototype

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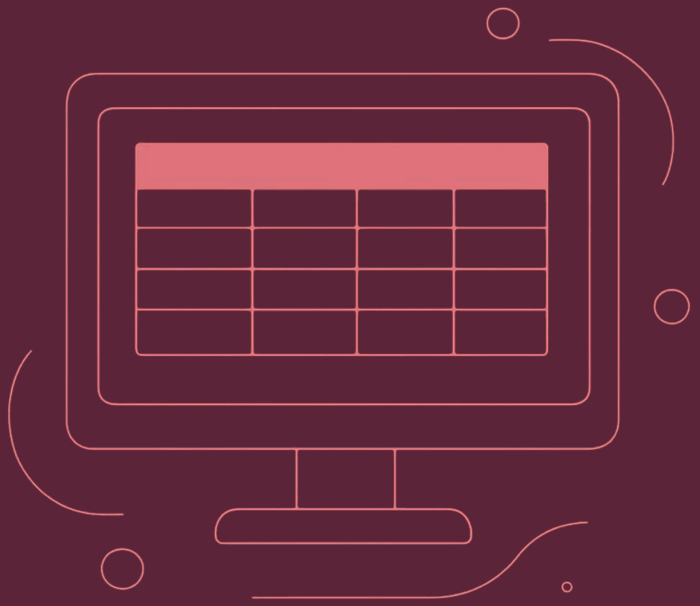
Dataset Overview



Key metrics

- Total transactions: 10,000
- Fraudulent: 500 (5%)
- Total volume: ₹1.78M
- Fraud loss: ₹828.79K (46.52% of volume)

Features analyzed: transaction type, merchant category, country, hour, amount, device risk score, IP risk score

Data quality: 0 missing values, 0 duplicates — clean and analysis-ready

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Methodology

01

SQL

Answered 10 business questions: fraud by type, category, country, hour, amount, risk score.

02

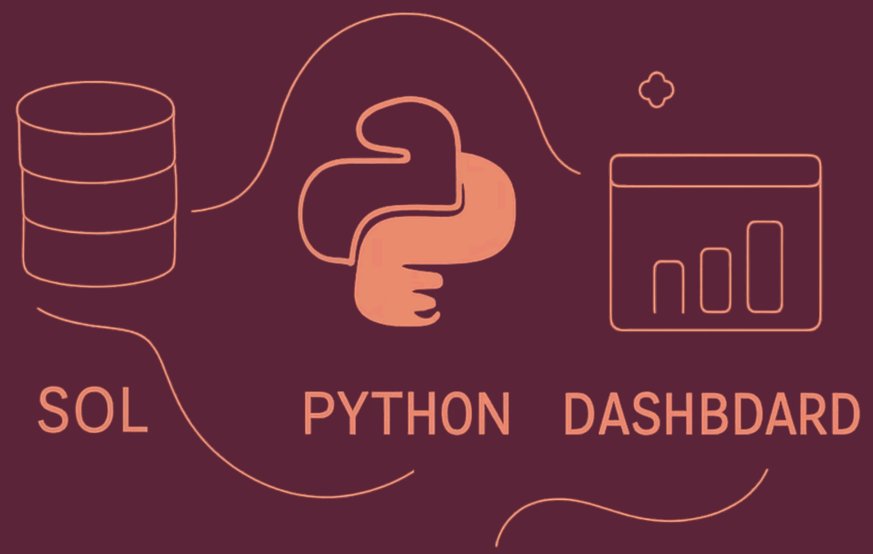
Python

EDA, correlation testing, statistical summaries, Matplotlib visualizations.

03

Power BI

Interactive dashboard for stakeholders and operational monitoring.



Pipeline ensured reproducibility: repeatable SQL queries, version-controlled notebooks, and a refreshable Power BI model.

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Where & When Fraud Happens

Top signals

- ATM transactions: highest fraud rate (5.46%).
- Night window (12–6 AM): 28.55% fraud rate — ~9x baseline risk.
- Country outlier: Nigeria flagged with 100% fraud rate in dataset.
- Merchant category: Clothing highest risk (5.50%).

Implication: Prioritize night-hour and ATM-origin transactions for automated screening.

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Amount & Risk Score Patterns

Key observations

- All transactions > ₹500 in the sample are fraudulent (100% rate).
- Mean fraudulent amount: ₹1,657.58 vs. legitimate: ₹100.28 — 16.5x difference.
- Device & IP risk scores correlate strongly with fraud (r = 0.87) — robust predictive signal.

Actionable threshold: treat transactions over ₹500 as high-priority for verification and blocking rules.

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Dashboard Walkthrough

Interactive Power BI prototype includes:

- Real-time KPI tiles: fraud rate, value at risk, alerts per hour
- Geographic fraud heat map with drill-through
- Transaction-type and merchant-category filters
- Value breakdown: fraud vs legitimate sums and counts

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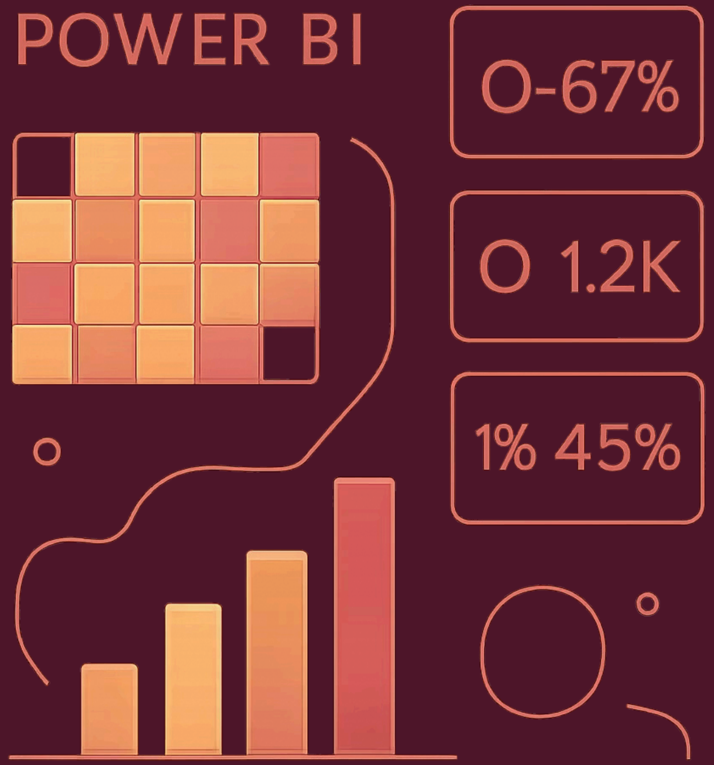

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Recommendations & Projected Impact

1

Restrict high-risk geographies

Immediate reduction in fraud originating from flagged countries.

2

Verify transactions > ₹500

Block or require step-up authentication for high-value items.

3

Monitor night-hour activity

Apply stricter rules during 12–6 AM window where risk is concentrated.

4

Deploy real-time risk scoring

Automated screening enables immediate blocking and alerting.

Projected impact

Metric	Estimate
Fraud reduction	60–70%
Estimated annual savings	₹497K – ₹580K

Estimates based on applying threshold-based rules and real-time scoring to the sampled distribution.

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Conclusion & Next Steps

What this project demonstrates

- End-to-end analytics workflow: SQL → Python → Power BI
- Ability to translate patterns into operational controls
- Strong predictive signals: amount thresholds and device/IP risk scores

Next steps

1. Build a supervised ML classifier using labeled sample and features identified
2. Expose a real-time scoring API for production screening
3. Pilot with live transaction stream and monitor KPIs

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